

YAROSLAV VOROPAEV

ML RESEARCHER · RL & LLM EFFICIENCY · REMOTE, PAPHOS, CYPRUS (EU)

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RESEARCH EXPERIENCE

[01 / 06]

ML Researcher — MWS AI Research Collaboration

Jun — Sep 2025

LLM · RL · NLP · inference optimization · quantization · pruning · distillation

Conducted LLM inference research in collaboration with MWS AI, focusing on quantization, pruning and distillation; developed a modular optimization pipeline and evaluated quality, latency and compression trade-offs. The work demonstrated up to **5x inference speedup** and **5x model compression** with INT4 quantization.

RESEARCH PROJECTS

[02 / 06]

act-lm — learned action vocabularies via RL

2026

RL · GRPO · Qwen2.5 0.5–1.5B · single RTX 3090 · github.com/zabojob/act-lm

Question: can a language model learn to mint its own operators instead of being handed a fixed action set? **Method:** trained with **GRPO** on Qwen2.5 (0.5–1.5B) on **one RTX 3090**, with ablations isolating the contribution of the learned vocabulary. **Result:** success rate **1.00** against **0.65** and **0.17** for the ablated variants.

Self-Play RL Agent — Kaggle Orbit Wars

2026

Imitation learning → self-play PPO · opponent modeling · **Silver Medal, 33rd / 4,729**

Method: bootstrapped from **imitation learning** on strong replays, then **self-play PPO** with a KL anchor against the imitation prior, then opponent modeling. **Lesson:** move-prediction accuracy is a misleading proxy — >90% agreement with a teacher policy still left win rate **5–10% below** it, so score the objective on outcomes.

Aspect-Based Sentiment Analysis for Course Reviews — Team ALT, AIJC 2024

2024

ML research & engineering · **2nd place overall** · Role: backend + model training

Goal: extract five aspects (theory, practice, instructor, technology, relevance) from Russian-language course reviews and classify sentiment per aspect. **Contribution:** after rejecting zero-shot, DBSCAN over TF-IDF/Word2Vec, and noisy LLM-augmentation, designed a **two-stage architecture** — fine-tuned xlm-roberta-large for aspect extraction, trained via pseudo-labelling over 100 000+ reviews; quantised Qwen2.5-72B served through vLLM on free Kaggle compute for per-aspect sentiment. Built the Flask backend: REST API, single-review and batch-CSV inference, web-UI integration. **Result:** macro-F1 = **0.639** — best across all team configurations; **2nd place** nationally.

OTHER EXPERIENCE

[03 / 06]

Agentic Workflows Developer — Bobr AI

May — Aug 2026

Agentic workflows · LLM · NLP · AI agents · RAG · workflow orchestration

Designed and implemented agentic workflows for automating multi-step LLM and NLP tasks: tool-calling pipelines, orchestration between steps, handling of failure modes in long-running agent runs, and **RAG systems over non-standard data**.

ML Instructor — Fit Predict

Aug 2026

Summer school with Letovo and HSE University · Machine Learning · Computer Vision · NLP

Designed and authored practical tasks and assignments in ML, computer vision and NLP; **reviewed solutions, checked correctness and quality**, and provided feedback to students — the same loop as evaluating model-generated code: define the task, decide what “correct” means, find where a solution silently breaks.

AWARDS & OLYMPIADS

[04 / 06]

MAJOR AWARDS

33rd / 4,729 · Silver Medal , Kaggle Orbit Wars	2026
2nd place , Yandex ML Challenge	2026
Bronze Medal , NeuroGolf 2026	2026
Prize Winner , All-Russian School Olympiad — Artificial Intelligence	2025
Finalist , All-Russian School Olympiad — Informatics, National Final	2025
Prize Winner ×2 , All-Russian School Olympiad — Information Security	2023, 2024
Silver Medalist (6th place), Russian National AI Olympiad — RSOSH Tier II	2025
2nd place team / 3rd place individual , National Technology Olympiad — AI track	2025
2nd place , AI International Junior Contest (AIJC) — Aspect-Based Sentiment Analysis	2024
12th place , North Eurasia Olympiad in Artificial Intelligence (NEOAI)	2025

ADDITIONAL DISTINCTIONS

Kaggle Expert · Codeforces Expert — rating 1845	2026
Prize Winner , T1.GenAI National AI Olympiad · Prize Winner , HSE Highest Probe Olympiad — Industrial Programming, RSOSH Tier II	2025
Speaker , CIPR (Digital Industry of Industrial Russia) Conference · 1st Sports Category in Competitive Programming	2024 – 2025
Prize Winner , Tact Smart Battle · Prize Winner , Kotlin Heroes	2024 – 2025

METHODS & TECHNICAL SKILLS

[05 / 06]

PROGRAMMING	Python (advanced), C++ (intermediate), JavaScript / TypeScript (intermediate), Kotlin & Java (basic), SQL (basic)
ML / NLP / DL	PyTorch, transformers, vLLM, ONNX, quantization & pruning, scikit-learn, CatBoost, LightGBM, implicit, pandas, NumPy, SciPy
REINFORCEMENT LEARNING	PPO, DQN, AWR / advantage-weighted regression, imitation learning, self-play, opponent modeling
METHODOLOGY	Leakage-aware validation (StratifiedKFold, GroupKFold, TimeSeriesSplit), ablations, baselines, blending & stacking, reproducibility
BACKEND	Python (Flask, FastAPI), REST API design, Docker (basic), Linux/bash
TOOLS	Git, Docker, Jupyter, LaTeX, Typst, Linux/bash
MATHEMATICS	Linear algebra, probability, statistics, calculus, discrete maths

EDUCATION & LANGUAGES

[06 / 06]

Neapolis University Paphos, Cyprus	Sep 2026 –
BSc Computer Science & Artificial Intelligence (CSAI)	
RUSSIAN	Native
ENGLISH	B2+ — verified by Password Skills Plus